Application of Lightweight YOLOv8 Variants for Defect Detection and Classification in Canned Food Packaging

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***Abstract*—There is a significant increase in health concerns regarding foodborne diseases in canned food products. Although packaging helps protect the food contents, defects in the packaging can increase the risk of contamination. Therefore, manual inspection must be carefully carried out before the products are released to consumers. However, manual inspection can be inaccurate and inefficient, as it requires a lot of time and labor, especially in high-volume mass food production. To address these issues, a real-time defect detection system for canned food packaging has been developed using the YOLOv8 object detection algorithm. The dataset for this project was collected from Teachable Machine and then pre-processed using the Roboflow platform. The research design for this project involves understanding requirements, data collection, model preparation, training, testing, and prototype development. The YOLOv8n variant was specifically selected for its lightweight architecture, which enables faster inference and lower computational requirements, and it achieved a mean average precision of 99.5% on a dataset of 1,292 images, with an overall accuracy of 96.7% during manual testing of the prototype. The prototype of the system is implemented as a web application. In the future, the project can expand the dataset to include more categories and types of defects in canned food packaging.**

***Keywords*—YOLOv8, packaging, canned, food, classification, deep learning**

I. Introduction

Food safety refers to the application of appropriate procedures throughout the food supply chain to prevent contamination and reduce foodborne illnesses among consumers [1]. There are 250 known dangers in food that can make people sick, including germs like Taenia solium, E. coli, and toxins like aflatoxins, which can lead to serious health issues such as epilepsy, kidney failure, or liver cancer [2]. One of the critical elements in ensuring food safety is packaging, which not only protects food from external contamination but also preserves its shelf life and quality [3]. In response to modern consumer demands for clean, minimally processed, and ready-to-eat products, the integrity of food packaging has become increasingly important. However, packaging materials remain vulnerable to defects such as punctures, leaks, and deformations caused by poor handling during manufacturing, transportation, or storage. These faults can compromise the food inside, leading to spoilage or contamination [4]. Furthermore, regulatory standards for exported goods emphasize the necessity for high-quality sealed packaging. As such, improving the reliability of food packaging systems is vital to maintaining product quality, reducing food waste, and protecting consumer health.

Despite the importance of packaging in safeguarding food products, current quality assurance processes still rely heavily on manual inspection. This method is time-consuming and prone to human error, especially in large-scale production settings. Workers may become fatigued after long hours and might overlook defective canned food. Defective packaging, even when minor, can result in air exposure and microbial contamination, directly compromising food safety. A recent case reported by [5] involved a major recall of 2.58 million pounds of canned food due to packaging issues that posed contamination risks. Such incidents not only threaten public health but also lead to significant economic losses and reputational damage for manufacturers. Furthermore, traditional image processing techniques, which involve mathematical operations and feature enhancement algorithms, have some limitations, such as limited parameter tuning and difficulty in classifying objects in complex environments. In contrast, deep learning automates the process and provides more accurate detection and classification, which can help strengthen quality control protocols in food manufacturing [6].

Therefore, the main aim of this project is to develop a real-time defect detection and classification system specifically for canned food packaging using lightweight deep learning models. The system can automatically identify defect categories: major, minor, and no defect on the food cans. The examples of defects in canned food products are shown in Fig. 1. This will help improve quality control in food manufacturing by reducing food waste and preventing the return of defective cans from consumers, thereby increasing consumer trust. Furthermore, the automated process will enhance operational efficiency, reduce labor workload, and maintain consistent product quality. The system also aids compliance with food safety regulations and industry standards, promoting ethical practices in the food manufacturing sector. This project contributes to food packaging safety by offering an intelligent solution that can enhance the effectiveness and reliability of defect detection in industrial environments.

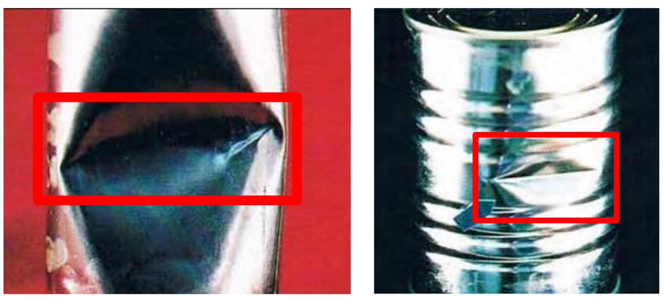

Fig. 1. Examples of major (left) and minor (right) defects in canned food products.

II. Scope of Project

This project primarily targets food production employees, particularly those involved in quality control operations within the canned food production area. Its main functionality is to detect and classify defects found on canned food packaging. The system is designed to categorize defects into three classes: no defect, minor defect, and major defect. Each category initially included 300 images, resulting in a total of 900 images collected from Teachable Machine (https://teachablemachine.withgoogle.com/). These images were then imported into the Roboflow platform for data preprocessing and data augmentation. As a result, the dataset increased to 1,292 images. The final dataset was split using a 70:20:10 ratio, producing 907 images for training, 255 for validation, and 130 for testing.

III. Related work

In recent years, object detection and deep learning models have become increasingly vital in enhancing food quality assurance and packaging inspection systems. A wide range of studies has explored the use of CNN and YOLO-based architectures to automate defect detection in food manufacturing, with varying results in terms of accuracy, dataset size, computational cost, and real-time applicability.

One major trend across these studies is the use of YOLO models (You Only Look Once), particularly YOLOv4, YOLOv5, and more recently YOLOv8, for real-time object detection of surface or packaging defects. For instance, models like YOLOv4 have demonstrated solid performance in classifying defect severity in canned food packaging, achieving up to 93% accuracy, with per-class precision ranging between 81.82% and 100% [7]. This was achieved using a relatively small dataset of 1000 manually annotated images, enhanced through contrast-based preprocessing. Although promising, the study noted some performance imbalance across defect classes, underscoring the importance of dataset balancing and defect variety in training robust models.

Comparatively, newer YOLO variants such as YOLOv5 and YOLOv8 have shown further improvements in both detection accuracy and training efficiency. For instance, YOLOv5, when applied to detect specific plastic surface defects like scratches or stains, yielded a mean average precision (mAP) of 73% and precision/recall scores of 0.83 after 1000 training epochs [8]. However, only one defect class was used for training, limiting the generalizability of the results. Meanwhile, the application of YOLOv8 in the metal manufacturing industry showcased the model's ability to scale across multiple variants (from lightweight YOLOv8n to deeper YOLOv8x), with the best-performing model (YOLOv8s) achieving a mAP of 90.6%, even when trained on rotated, augmented data [9]. This study also demonstrated the advantage of newer YOLO versions in maintaining high detection performance while reducing model complexity and training time.

On the other hand, several works have focused on CNN-based classification models such as ResNet, DenseNet, GoogleNet, and ShuffleNet. These models, while not as fast as YOLO for real-time detection, are often preferred for image classification tasks involving food quality. For example, CNN architectures were tested on fruit freshness datasets involving over 40,000 images across different fruits and freshness levels. DenseNet, in particular, stood out with precision as high as 97.97%, highlighting its effectiveness in multi-class classification tasks involving subtle visual cues [10]. However, such models required substantial data preprocessing and were more susceptible to overfitting, particularly in the presence of class imbalance.

A different study comparing five CNN models for defect classification in canned food; ResNet18, ResNet50, ResNet101, GoogleNet, and ShuffleNet showed that deeper networks like ResNet101 achieved the highest test accuracy (95.56%) [11]. Yet, training these models without GPU support introduced significant computational delays, raising concerns about their practicality in real-time industrial settings. Despite high classification accuracy, CNNs were largely limited to post-capture inspection rather than live detection, unlike YOLO models which integrate real-time object localization.

In terms of dataset preparation, most studies relied on manual annotation and basic data augmentation techniques, such as rotation or contrast adjustment [12], to improve model generalization. Tools like Roboflow and CVAT were commonly used for labeling. However, small dataset sizes remained a common limitation across multiple studies, which often led to biased predictions, especially for rare defect classes. Only a few studies, such as the fruit freshness detection one, worked with sufficiently large datasets (more than 40,000 images), enabling deeper model training with more stable results.

Overall, while CNNs tend to excel in classification accuracy, YOLO-based models; especially in their latest versions; strike a better balance between speed, accuracy, and real-time processing capabilities, making them more suited for practical deployment in industrial food quality assurance systems. Still, both model types face challenges such as limited dataset diversity, lack of standard benchmarks, and hardware dependency, especially in resource-constrained environments.

Thus, to further improve the detection in canned food packaging, future research should focus on balancing datasets, exploring lightweight model variants that can be integrated with portable devices, and incorporating multi-class defect detection that can handle both minor and major defects that can perform well in real-world conditions. This current study aligns with that direction by exploring YOLOv8 lightweight variants and including more defect classes, providing further insight into their potential for real-time defect detection in canned food packaging.

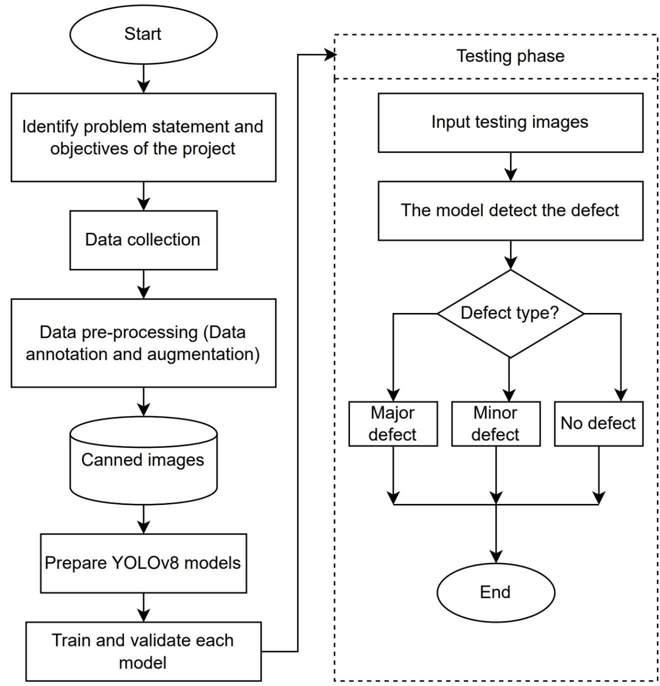

Fig. 2. Research design of the proposed system.

IV. Methods

The research design of the project is shown in Fig. 2. It involves identifying the problem and objectives of the project, as well as data collection, preprocessing, data annotation, training, and testing. The proposed system uses the deep learning model YOLOv8n for object detection. The performance of YOLOv8 variants; YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x were tested and compared to identify the most suitable model for the prototype.

A. Data collection

The sources of dataset has been mentioned in Section II. The dataset then categorized into three groups: Major Defect, Minor Defect and No Defect.

B. Data preprocessing and data annotation

The collected images were uploaded into Roboflow, a platform used for important preprocessing tasks in computer vision projects. Roboflow offers simple tools for labeling, cleaning, and improving images, which help prepare the dataset for training deep learning models [13].

The initial step involved manually annotating each image based on the specified defect classes, as depicted in Fig. 3. This involved drawing boxes around the damaged parts of the cans, making sure each image was correctly labeled based on how serious the defect was. This step is important because it teaches the model how to find and tell apart different types of packaging defects.

After all the images have been labeled, they are oriented in the same direction to ensure consistency in the dataset. The images are also resized to a standardized size of 640x640 pixels, which is suitable for input into YOLOv8 [14]. Next, to increase the dataset size, data augmentation is performed by creating mirror images of each image in the dataset [15].

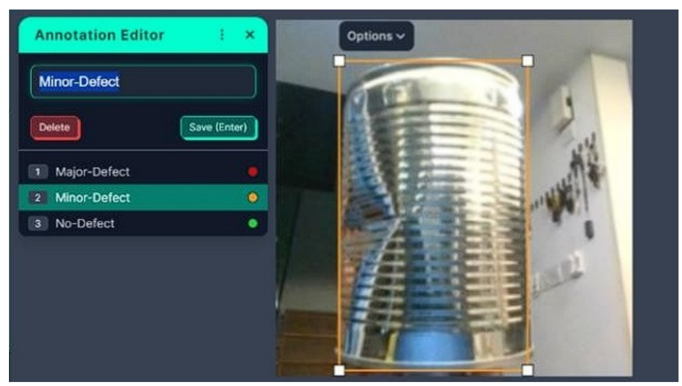

Fig. 3. Example of annotation labeling using Roboflow.

C. The architecture of algorithm

The architecture of YOLOv8 deep learning model is shown in Fig. 4. YOLOv8 builds on the CSPDarknet-53 backbone from YOLOv5 but replaces the C3 module with a new C2f module. This change improves how the model learns by allowing better gradient flow and easier adjustment of channel numbers. The model is also made more lightweight by optimizing the number of channels used, making it faster and more efficient [16].

In the head part of the model, YOLOv8 introduces two main features. One is the decoupled head, which separates the tasks of object classification and object detection. This helps the model focus on each task more clearly. The other feature is that it no longer uses anchor boxes, which were required in earlier versions. This anchor-free design makes training and detecting objects more flexible and accurate.

YOLOv8 also improves how it calculates errors during training. Instead of using the traditional Intersection over Union (IoU), it now uses Distribution Focal Loss (DFL), which helps the model better understand the exact location of objects in the image.

YOLOv8 comes in several variants, such as YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large) [17]. These versions offer different levels of speed and accuracy depending on the computing power available. For example, YOLOv8n is designed for very fast detection with low resource usage, meaning it requires less memory and processing power to run efficiently on smaller devices such as laptops or embedded systems [18]. On the other hand, YOLOv8x provides higher accuracy but needs more computing power, such as high-performance GPUs [19].

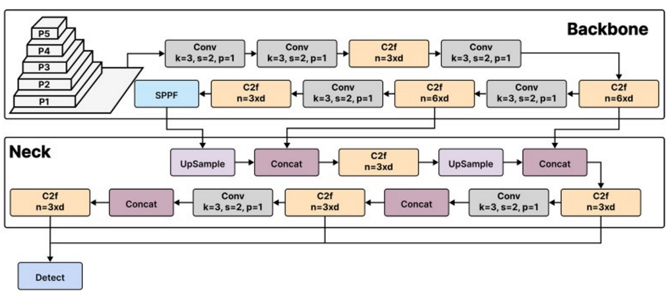

Fig. 4. The architecture of YOLOv8 deep learning model [20].

The YOLOv8n configuration is optimized for efficiency by using a depth multiplier of 0.33 and a width multiplier of 0.25, resulting in a lightweight architecture with approximately 3.2 million parameters and 8.7 GFLOPs. The model operates on a default input resolution of 640×640 pixels. It maintains the core YOLOv8 structure, including the C2f-enhanced CSP backbone, PAN neck, and decoupled head for classification and localization. These settings enable real-time object detection with minimal latency.

The training process of YOLOv8n involves optimizing a combination of loss functions. The bounding box regression loss is calculated using the Complete Intersection over Union (CIoU), which considers the overlap area, center point distance, and aspect ratio between the predicted box B_{pred} and the ground truth box B_{gt}:

$$\mathcal{L}_{\text{bbox}} = 1 - \text{CIoU}(B_{\text{pred}}, B_{\text{gt}}) \tag{1}$$

The classification loss is computed using binary cross-entropy for each class i, where C is the total number of object classes, $y_i \in \{0, 1\}$ is the ground truth label, and p_i is the predicted probability for class i:

$$\mathcal{L}_{\text{cls}} = \frac{1}{C} \sum_{i=1}^{C} \left[-y_i \log(p_i) - (1 - y_i) \log(1 - p_i) \right] \tag{2}$$

These loss components are combined with the objectness loss to form the total loss used to train the model, improving its accuracy in object detection and classification tasks.

D. The system

The system starts with a homepage where the user can click the camera button to activate real-time camera detection. If an object, such as a can, is detected, the system will then identify and classify the condition of the can. The system will show the result to the user, displaying whether the can is categorized as having a major defect, minor defect, or no defect at all. In Fig. 5 showed the detection page of the proposed system includes instructions for the user on how to use the system.

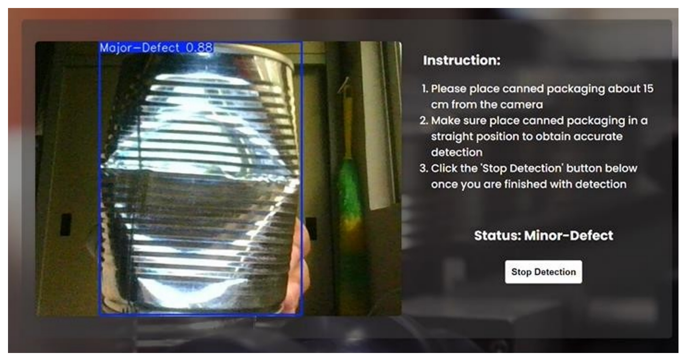

Fig. 5. The detection page of the proposed system includes instructions for the user.

V. Results and Testing

Two types of tests were conducted: evaluation metric testing and manual accuracy testing.

A. Evaluation metric testing

The performance of the YOLOv8 model is evaluated through evaluation metric testing to measure its object detection precision. Two main metrics are used: mean Average Precision (mAP) and inference speed [21]. The mAP indicates how accurately the model detects and classifies objects across different categories, where a higher mAP reflects better detection performance. Inference speed refers to the time taken by the model to process an image and produce a prediction, measured in milliseconds per image. The formulas are shown in Eq. 3 and Eq. 4, respectively.

In this section, the focus is primarily on the results of YOLOv8n.

$$mAP = \frac{1}{n} * sum(AP) \tag{3}$$

$$\text{Inference Speed}_{\text{ms/image}} = \frac{\text{Total Inference Time (ms)}}{\text{Number of Images}} \tag{4}$$

The model was validated using 255 images: 54 with major defects, 54 with minor defects, and 147 with no defects. It was then tested using 130 images: 25 with major defects, 34 with minor defects, and 71 with no defects.

TABLE I
VALIDATION AND TESTING RESULTS OF YOLOV8N

Defect Class	**Validation**		**Testing**	
	Instances	AP (%)	Instances	AP (%)
Major	54	99.5	25	99.5
Minor	54	99.5	34	99.5
None	147	99.5	71	99.5
Total	255	298.5	130	298.5
mAP@0.50	99.5		99.5	
Inference Speed	5.8 ms/image		8.3 ms/image	

Table I presents a comparison of the validation and testing results, including detection performance and inference speed.

The model achieved an Average Precision (AP) of 99.5% for each class and a mean Average Precision (mAP@0.5) of 99.5% in both the validation and testing phases, demonstrating consistent and strong performance. However, the identical scores across variants and datasets may suggest that the task is relatively simple, or that the metric (mAP@0.5) is not sensitive enough to capture differences in detection precision. Future evaluations will include mAP@0.5-0.95 to better distinguish model?s generalization ability. However, a difference in inference time was observed: 5.8 milliseconds per image during validation and 8.3 milliseconds per image during testing. The differences in results may be due to variations in image complexity or system performance during evaluation. Despite this, the results confirm the YOLOv8n model's robustness and reliability for detection tasks, indicating that it is not overfitting and can perform well in real-time scenarios.

1) Analysis of the performance results of YOLOv8 variants: As mentioned in Section IV, this project used different YOLOv8 variants, including YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x, to identify the highest-performing model suitable for integration into the prototype.

As shown in Table II, all variants maintain an impressive mean average precision (mAP) of 99.5%, reflecting their precision in defect detection across different scenarios, as indicated by their precision-recall curves.

In terms of inference time, YOLOv8n stands out with the smallest model size of 30.06 million (M) parameters, which contributes to its fastest inference speed of just 5.7 milliseconds (ms) per image among the variants. These features make YOLOv8n especially suitable for real-time applications on web platforms. Its compact size means it requires less memory and computational power, enabling faster loading and lower latency [22], [18]. The quick inference time allows nearly instantaneous image processing, which is essential for maintaining high responsiveness in real-time defect detection.

On the other hand, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x, although accurate, have larger model sizes and slower inference speeds. This can result in increased latency and slower processing, making them less ideal for real-time scenarios. Therefore, YOLOv8n is the best choice for integration into the system, balancing accuracy and efficiency to meet the project's demands for fast and reliable defect detection in a web application.

TABLE II
COMPARISON OF PERFORMANCE RESULTS OF YOLOV8 VARIANTS

Model	**Model Size (M parameters)**	**Inference Time (ms)**	**mAP (%)**
YOLOv8n	30.06	5.8	99.5
YOLOv8s	111.26	11.3	99.5
YOLOv8m	258.41	22.4	99.5
YOLOv8l	436.08	36.7	99.5
YOLOv8x	681.26	67.1	99.5

B. Manual Accuracy Testing

This section presents the manual accuracy results of the YOLOv8n model in detecting and classifying defects in canned packaging during real-time testing. During the test, actual cans; either with defects or without; were placed in front of the live camera to evaluate whether the system could accurately detect and classify the defects based on their severity. The accuracy is calculated using Eq. 5.

$$\text{Accuracy }(\%) = \left(\frac{\text{Number of correct predictions}}{\text{Total number of predictions}}\right) \times 100 \tag{5}$$

As shown in Table III, a total of 30 real-time test images were used, with 10 images for each defect class. The results show all classes achieved 100% accuracy except for the "No-Defect"class, where one image was mistakenly classified as "Minor-Defect", resulting in 90% accuracy for that class. Some misclassifications occurred, possibly due to inconsistent lighting or visual ambiguity in the defect presentation. Overall, the YOLOv8n model achieved a 96.7% accuracy rate. These results demonstrate that the system integrated with YOLOv8 can detect and classify defects in canned packaging in real-time.

TABLE III
REAL-TIME TESTING ACCURACY BY DEFECT LEVEL

Defect Level	**Total Images**	**Correct Classifications**	**Accuracy (%)**
Major-Defect	10	10	100
Minor-Defect	10	10	100
No-Defect	10	9	90
Total	**30**	**29**	**96.7**

VI. LIMITATION OF THE PROJECT

This project primarily focuses on detecting defects on tall, cylindrical canned food packaging. The dataset is also limited to a single packaging color; silver-grey. Additionally, the cans do not have any plastic labels, which is only of bare-metal surfaces. Furthermore, detecting defects is challenging due to inconsistent lighting conditions. For instance, the system may detect a can as having no defects, even though it visually resembles a can with minor defects.

VII. Future Work and Recommendations

For future work, the project will incorporate different images of cans and defect types, including a variety of can colors and packaging materials. Additionally, the model's performance can be enhanced by integrating optimizers [23]. Furthermore, exploring the latest lightweight deep learning models, such as YOLOv11 and other deep learning architectures, may provide additional improvements in system detection performance [24]. Using less controlled lighting environments is also essential to ensure consistent detection accuracy. The testing setup can be expanded to real food manufacturing settings. Finally, the system can be upgraded with a dashboard visualization and real-time notifications and report to monitor the total number of defective cans detected in a day.

VIII. Conclusion

In conclusion, this project addresses critical concerns related to food contamination caused by defects in canned food packaging, which, if overlooked, can increase health risks, reduce shelf life, and lead to food waste. To mitigate these risks, an automated real-time defect detection system was developed using the lightweight YOLOv8 deep learning model. Several variants of YOLOv8 were evaluated, and YOLOv8n was selected for its fast inference time and reliable performance, making it suitable for web-based deployment. The system's performance was validated using a testing dataset, achieving a mean average precision (mAP@0.5) of 99.5% and an overall accuracy of 96.7%. The resulting prototype demonstrates strong potential to enhance food safety and operational efficiency in real-world food manufacturing environments.

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